OHIO SCIENTIFIC

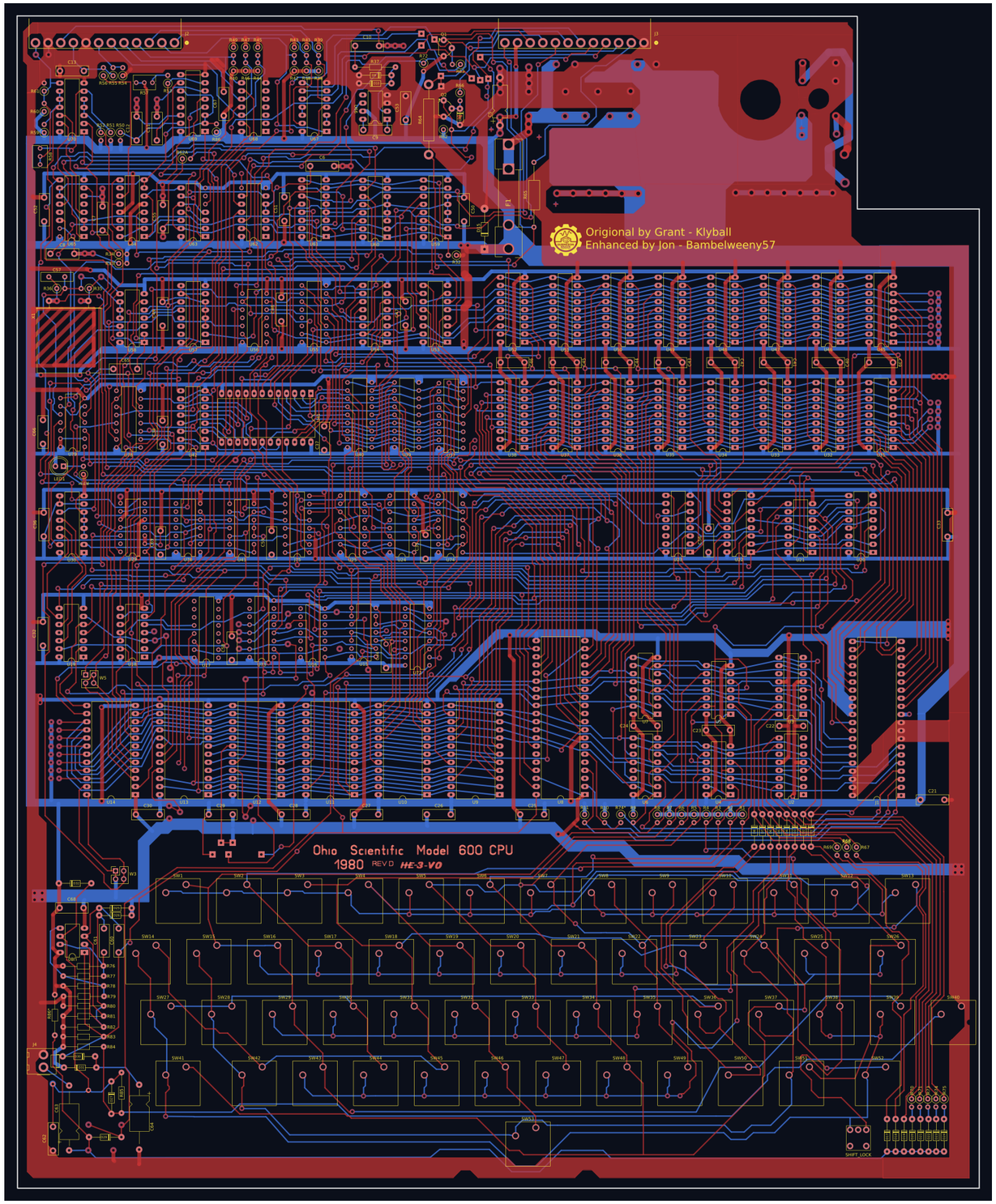
Superboard II

Model 600 – Revision D

Cherry MX keyboard edition

Schematic sheets 1–5 (vector traces of the 1980 originals)
Component placement drawing
Notes on the KiCad 10 project

Original design: Ohio Scientific, schematic dated 28 Aug 1980
Original PCB copper reconstruction: Grant – Klyball
Enhanced edition and KiCad project: Jon – Bambelweeny57
Submeson Brain Corporation
Documentation rev 1.0 – November 2026



All copper layers and silkscreen, viewed from the top (rev 1.0 board)

Foreword

This booklet goes with the KiCad 10 project for a reproduction of the Ohio Scientific Superboard II, Model 600 revision D. It collects the five original OSI schematic sheets, a placement drawing of the new board, and notes on everything that was renamed, added or changed so that the design works as a native KiCad project.

The board

The copper is based on Grant (Klyball)'s reconstruction of the original board from Gerber artwork. This edition, by Jon (Bambelweeny57), keeps that layout and adds:

- Cherry MX keyswitches in place of the original keyboard.
- ROM sockets rewired for 27C16 EPROMs or 2816 EEPROMs (monitor and BASIC), and a 2716-type character generator.
- Pin headers in place of the original wire-jumper blocks, so options are set with jumper links.
- A separate +5V_IN supply entry, so the fuse F1 sits ahead of the reverse-polarity diode D15 and C5.
- Exposed-copper lettering and the Submeson Brain logo.

Everything else follows the original design, and the netlist was checked pin by pin against the 1980 sheets.

The KiCad project

The schematic has five hierarchical sheets that follow the five OSI pages. Logic ICs are drawn as gates, flip-flops and one-shots, each on the sheet where OSI drew it, and each IC's power pins sit with its decoupling capacitor. Connections are made with net labels; drawn wires are still being added. The schematic netlist matches the PCB exactly (269 nets).

Nets carry the names used on the OSI sheets wherever the original gives one (PHI2, /MCS, Tx CLOCK, C0–C15 and so on). Where the original has no name, the net was named after its function, for example /RAM_0000 for the RAM chip select at $0000.

About the schematic pages

Pages 3 to 7 are vector traces of the original scanned drawings. They were cleaned up and converted to outlines automatically, so they print and zoom sharply, but they are not redrawn: anything hard to read on the scan is still hard to read here. Where this board differs from the 1980 drawings, the board and the KiCad schematic take precedence. Each page lists those differences in the box at the bottom.

Reading net names

Active-low signals are shown with a leading slash in this booklet, for example /MCS. In KiCad they are written ~{MCS} and drawn with an overbar.

Thanks

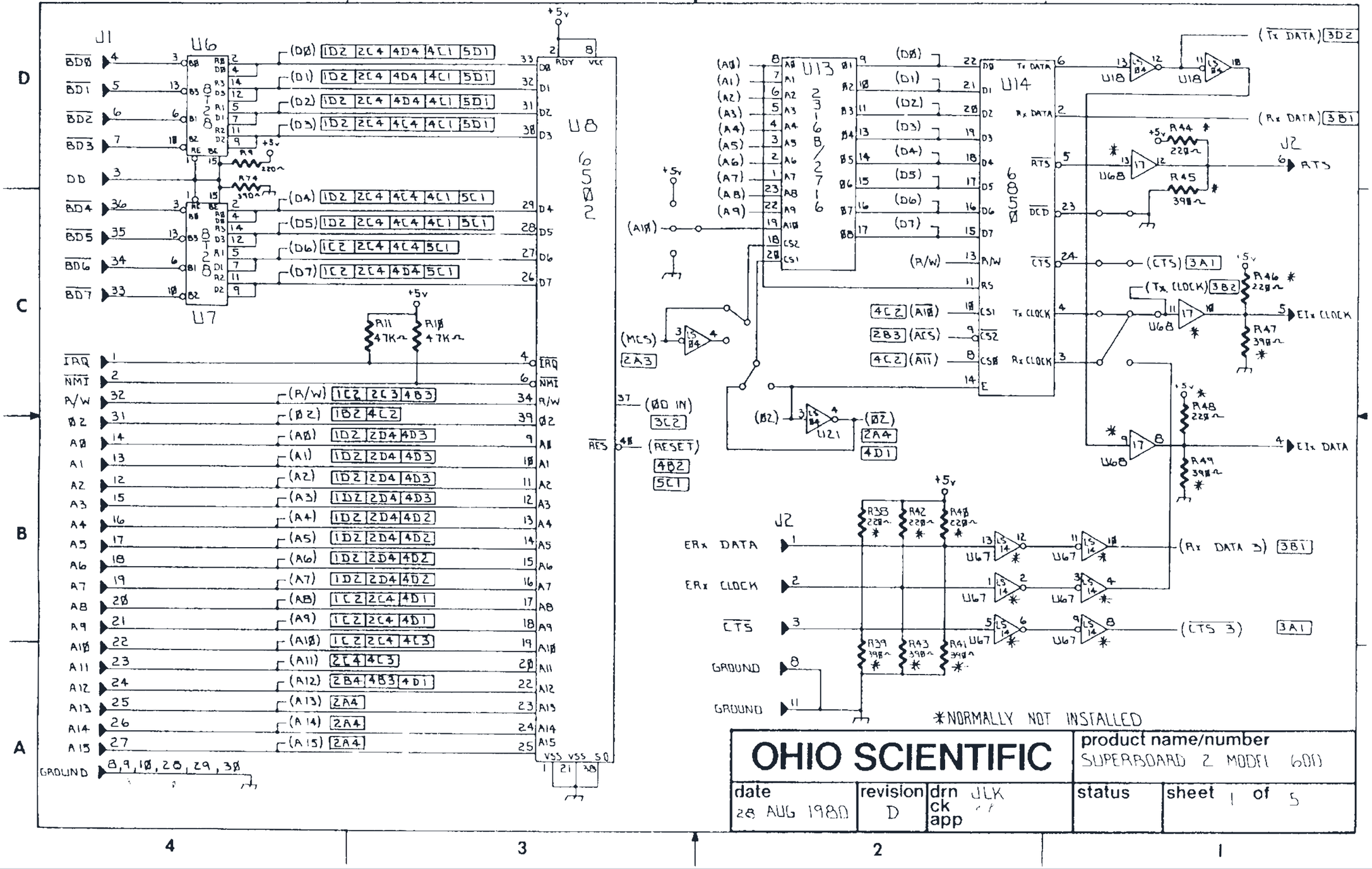
To Ohio Scientific for the original design and documentation, and to Grant (Klyball) for the copper reconstruction this board is built on.

Contents

Page	Content
1	Cover
2	Foreword
3	Sheet 1 – CPU, bus buffers, monitor ROM, ACIA, external serial
4	Sheet 2 – program RAM, BASIC ROMs, address decode
5	Sheet 3 – oscillator, video timing, baud clock, cassette, RS-232
6	Sheet 4 – video RAM, colour RAM, character generator, control register
7	Sheet 5 – keyboard, sound DAC, reset, power
8	Component placement, top side
9	Notes: reference designators, part values, build notes, net names

Project files

File	Contents
OSI 600D Superboard II Cherry Solder Mask.kicad_sch	Top-level schematic with the five sheets
Sheet1_CPU_Bus_ACIA.kicad_sch	Sheet 1
Sheet2_Memory_Decode.kicad_sch	Sheet 2
Sheet3_Timing_Serial_Cassette.kicad_sch	Sheet 3
Sheet4_Video.kicad_sch	Sheet 4
Sheet5_Keyboard_Reset_Power.kicad_sch	Sheet 5
OSI 600D Superboard II Cherry Solder Mask.kicad_pcb	Printed circuit board



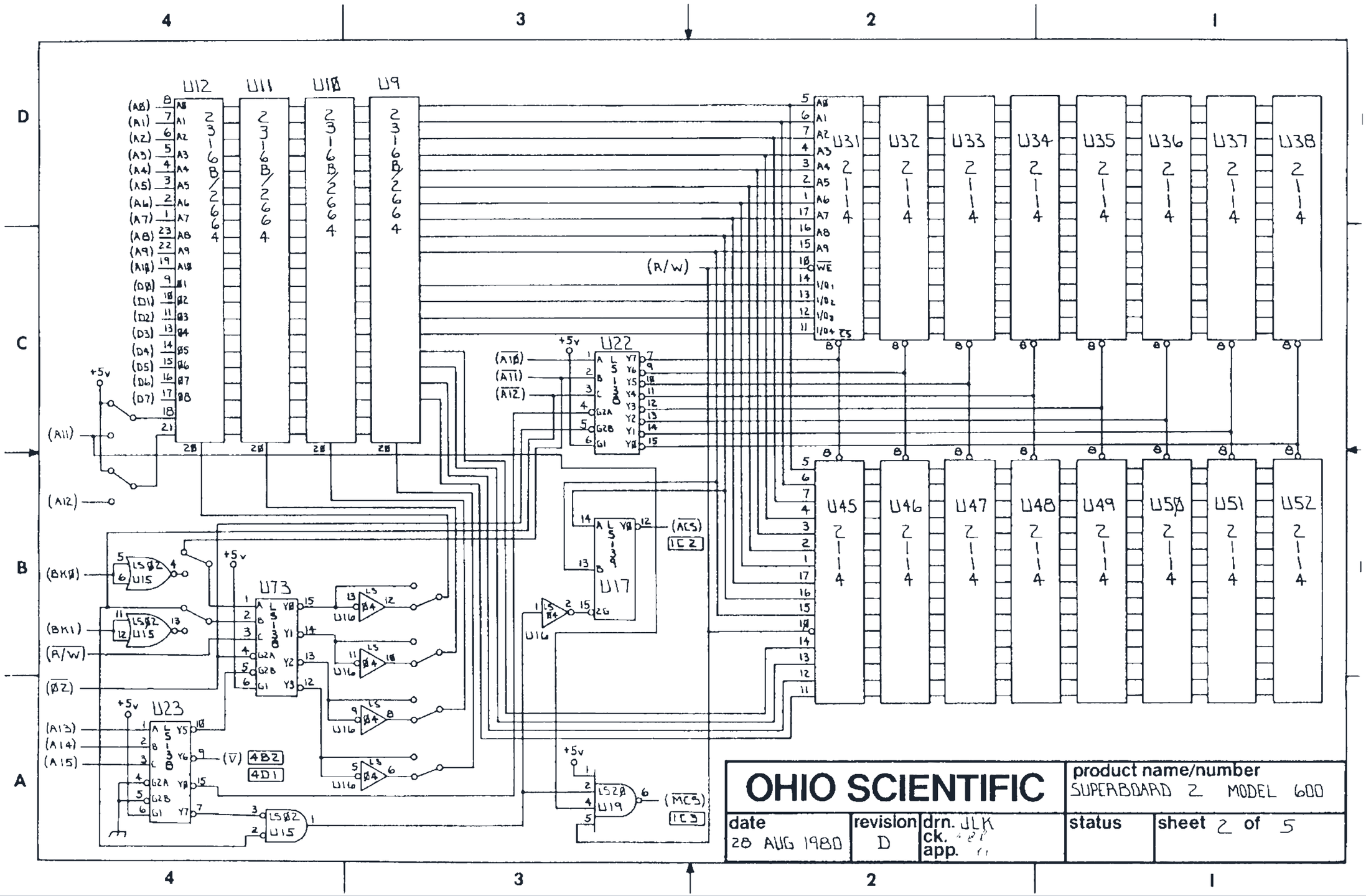
On this board

U13 monitor ROM is a 27C16 EPROM or 2816 EEPROM: pin 18 (CE) = /MCS, pin 20 (OE) = /PHI2, pin 21 = +5 V. The A10/CS jumpers around U13 are not used.

R74 (390 Ω, DD to ground) is R88 on the board and is not fitted (R9 holds DD high on its own).

The jumper options (CTS and receive clock) are on pin headers. U67 outputs CTS 3 and Rx DATA 3 go to jumper holes beside J3.

Vector trace of the original OSI scan



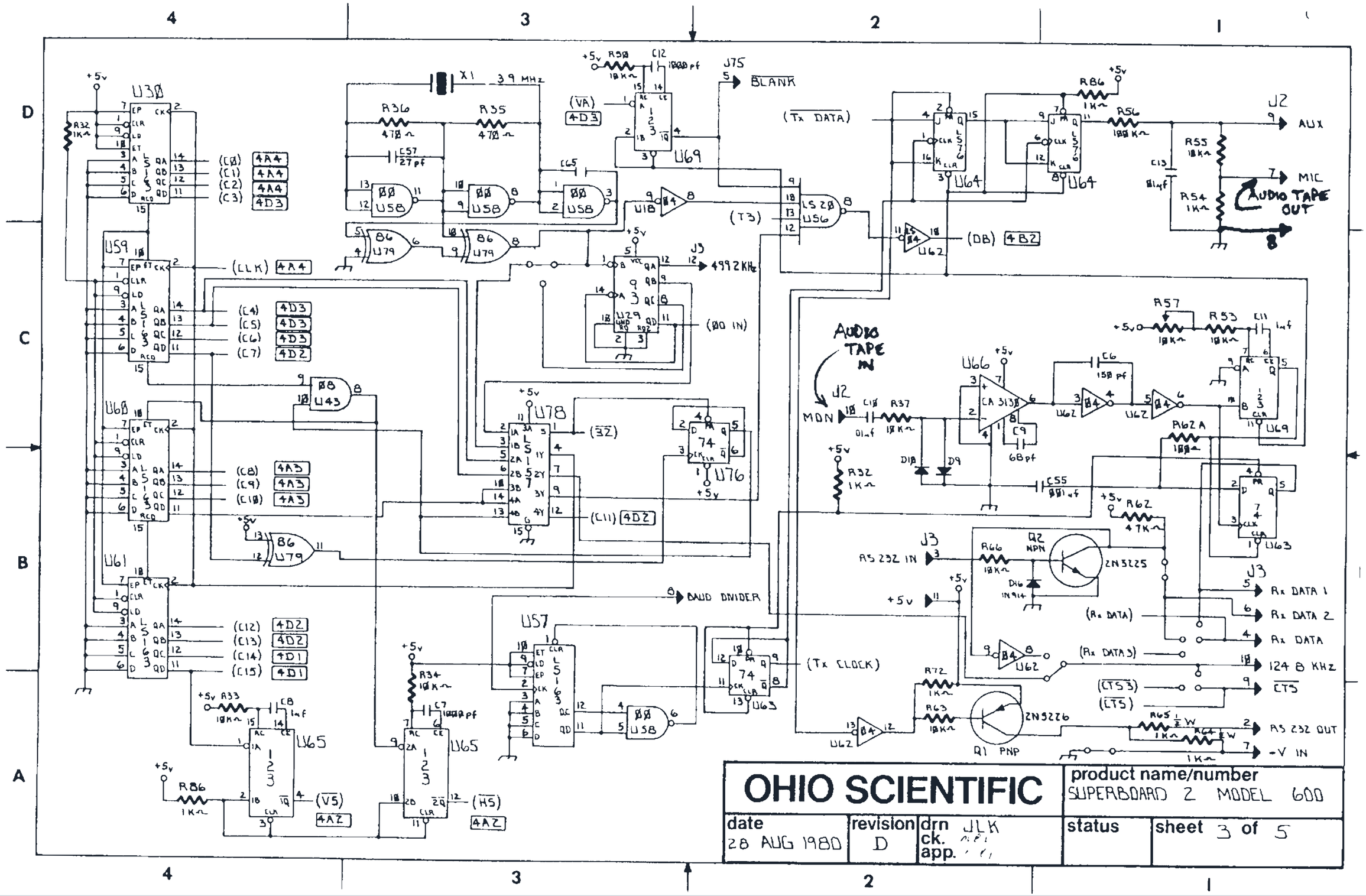
On this board

BASIC ROMs U9–U12 are 27C16/2816: pin 18 (CE) tied to GND, pin 20 (OE) driven from U73, pin 21 = +5 V.

U73 inputs A and B are wired directly to /A11 and /A12, so the BK0/BK1 bank-select jumpers are not used. The ROMs sit at $A000, $A800, $B000 and $B800.

U17 is a 74LS139 (dual 2-to-4 decoder).

Vector trace of the original OSI scan



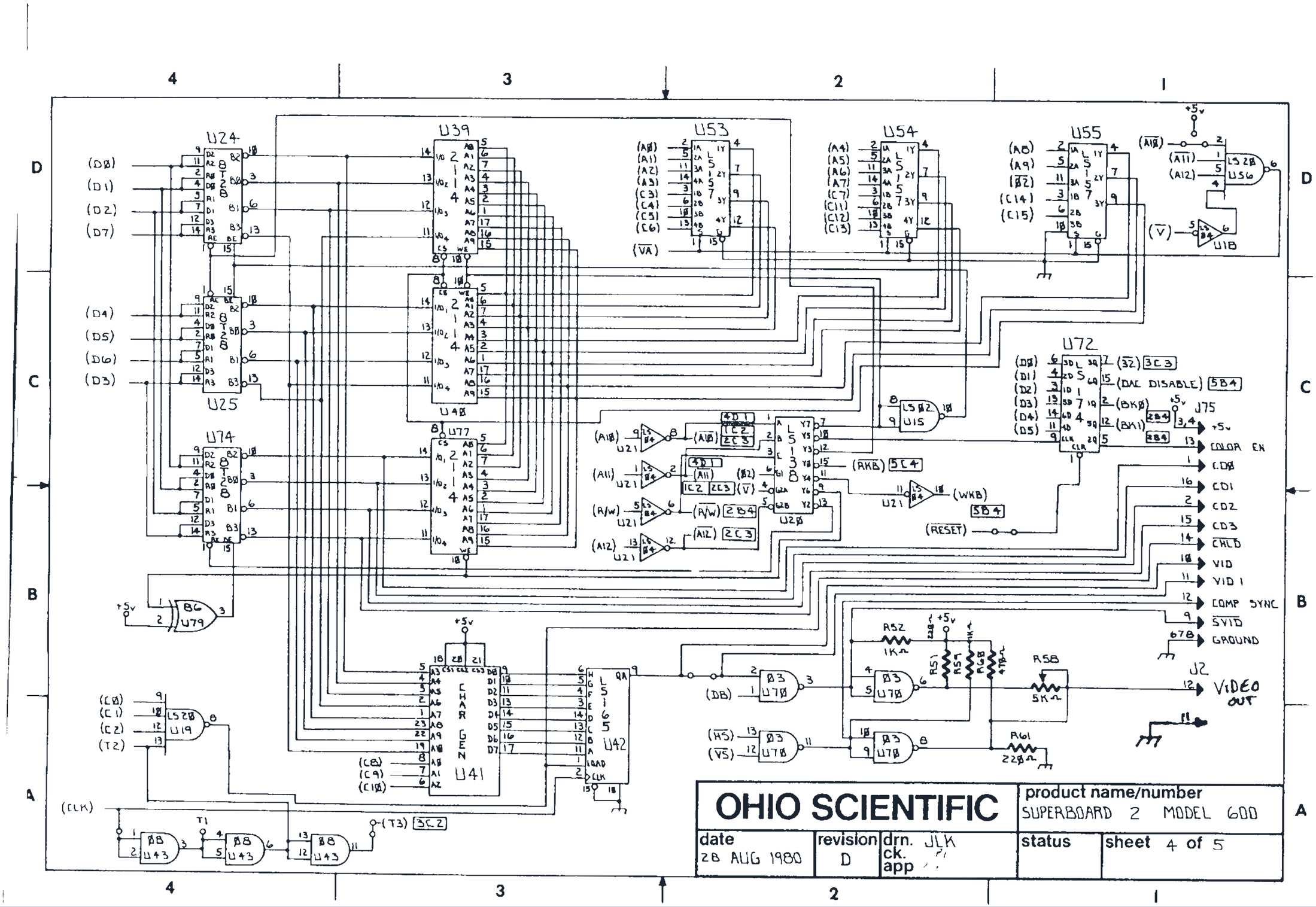
On this board

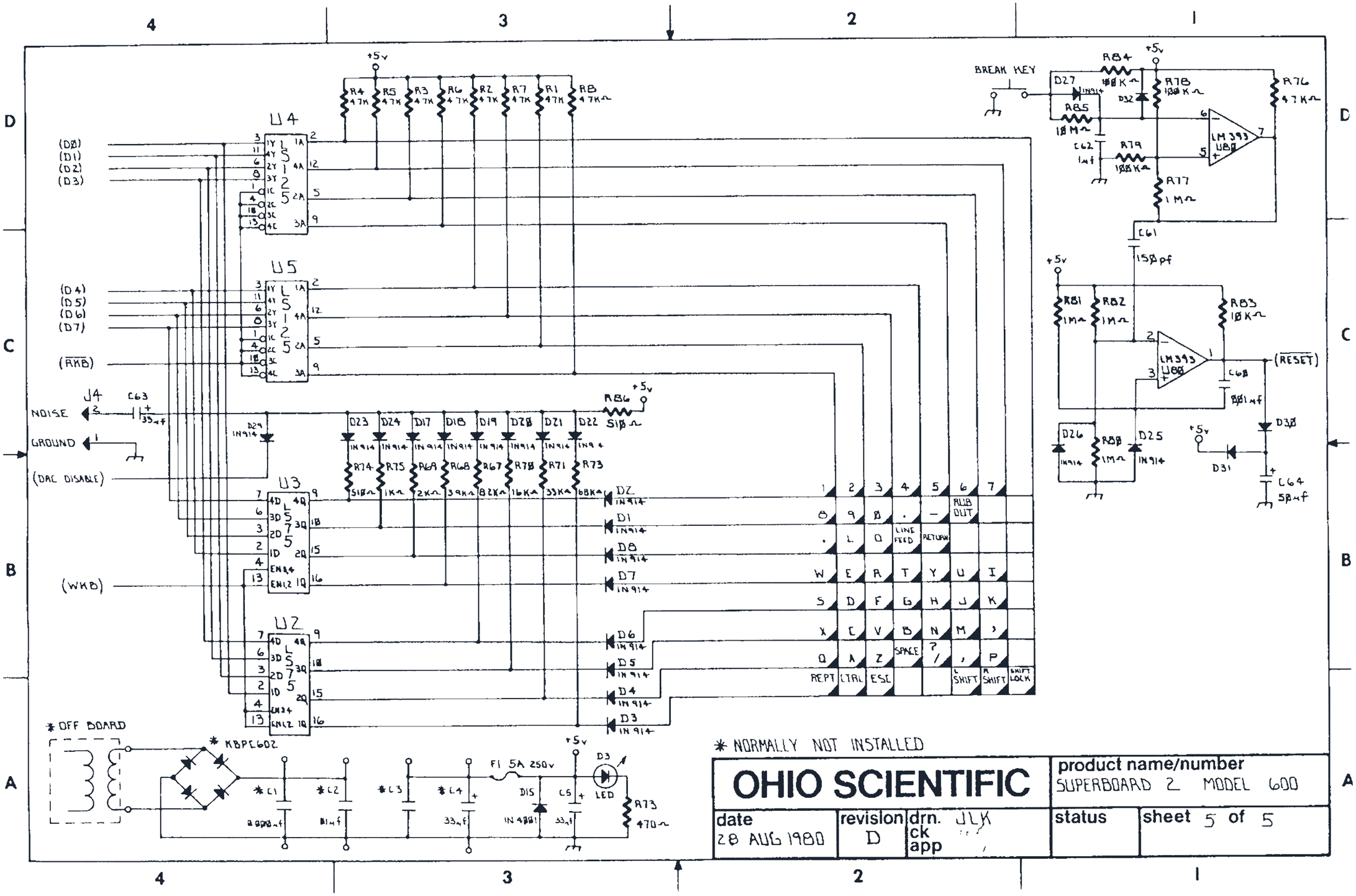
R62A is R90 on the board. R86 on this sheet (1 kΩ pull-up) keeps its number.

U29 input B is wired to the U79 clock doubler (the jumper option is not fitted). J3 pins 10 and 12 carry 124.8 kHz and 499.2 kHz.

J3 pin 7 (–V IN) is tied to GND, so the RS-232 output swings 0 to +5 V.

Vector trace of the original OSI scan


U24
U25
U74
U39
U48
U77
U53
U54
U55
U72
U75
U18
U15
U79
U41
U42
U43
U70
U71
U19
U21
U20
U56
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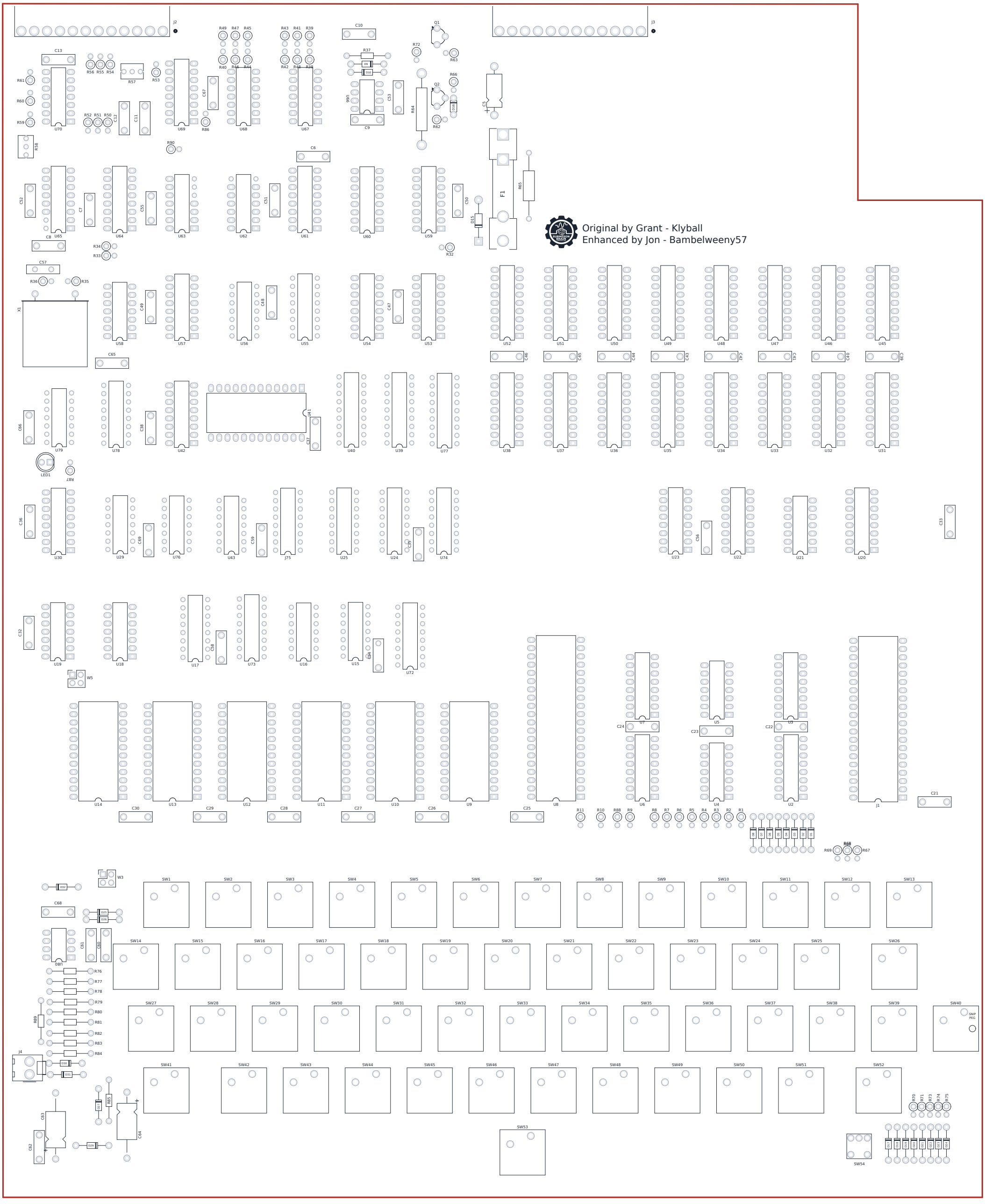
On this board

Keys are Cherry MX switches SW1–SW53; Break is SW40. Shift lock is SW54, which is wired in parallel with SW38 at the same matrix position – fit one of them.

R86 on this sheet (510 Ω DAC pull-up) is R89. D3 (LED) and R73 (470 Ω) in the power section are LED1 and R87. R85 is 10 MΩ.

Power enters at +5V_IN and passes through fuse F1 before D15 and C5. On SW40, snip the right-hand plastic peg (marked on the silkscreen).

Vector trace of the original OSI scan



Board outline 301.5 × 367.3 mm. Drawing scale about 1:1.09. Grey: pads; white: holes; black: silkscreen; red: board edge. Zoom in for reference designators.

Notes

Reference designators

KiCad only treats a reference as annotated if it ends in a number, and every reference must be unique. These parts were renumbered or added:

On the OSI sheets	On the board	What it is
R74 (sheet 1)	R88	390 Ω DD pull-down. Connected to GND, normally not fitted
R62A (sheet 3)	R90	100 Ω cassette input resistor (KiCad needs a numeric reference)
R86 (sheet 5)	R89	510 Ω sound DAC pull-up. The 1 kΩ R86 on sheet 3 keeps its number
D3 LED (sheet 5)	LED1	Power LED. D3 is the keyboard diode on the same sheet
R73 470 Ω (sheet 5)	R87	LED resistor. R73 is the 68 kΩ DAC resistor
Shift lock (sheet 5)	SW54	Shift-lock switch, in parallel with SW38
Keyboard	SW1–SW53	Cherry MX switches. SW40 is Break
–	H1–H21	Mounting and stabiliser holes (board only, from the original drill data)
–	G***	Submeson Brain logo (board only)

Part values

Part	Value used	Note
U17	74LS139	Address decode, sheet 2
U54	74LS157	Video address multiplexer, sheet 4
U9–U13	27C16 / 2816	Monitor and BASIC ROMs (originally 2316B/2716)
U41	2716 type	Character generator with OSI font
R85	10 MΩ	Break-key timing, as on sheet 5
C5	47 µF	33 µF on sheet 5; larger value fitted
R88	do not fit	See above

Building the board

- Supply 5 V at the +5V_IN holes; the rest of the board is fed through fuse F1.
- Do not fit R88.
- On the Break key (SW40), snip off the right-hand plastic peg before fitting; the hole was removed because it cut the ground strip.
- Fit only one of SW38 and SW54 (shift lock).
- ROMs: use 27C16 or 2816 parts of 450 ns or faster. The character generator needs the OSI character set in 2716 layout.
- The 8T28 bus transceivers (U6, U7, U24, U25, U74) are obsolete and need new-old-stock parts; the 8T26 is inverting and will not work.
- Bring the board up in stages: check Vcc to GND resistance, then power only, then CPU, monitor ROM and RAM, then video.

Net names

OSI names are kept where the original sheets give one. Rows marked (none) are nets the original sheets leave unlabelled; those names were added for this project.

OSI label	KiCad net	Notes
Ø2, /Ø2, Ø0 IN	PHI2, /PHI2, PHI0	CPU clocks
MCS, ACS	/MCS, /ACS	Monitor ROM and ACIA selects
V, VA	/V, /VA, V	Video RAM decode
RKB, WKB	/RKB, WKB	Keyboard read and write
BK0, BK1, 32	BK0, BK1, MODE32	Control register outputs
Tx DATA, Tx CLOCK	TXDATA, /TXDATA, Tx_CLOCK	ACIA transmit side
ERx / ETx DATA, CLOCK	ERx_DATA, ERx_CLOCK, ETx_DATA, ETx_CLOCK	External serial on J2
Rx DATA 1, 2, 3; CTS, CTS 3	Rx_DATA_1..3, /CTS, /CTS_3	Receive and CTS options
124.8 KHz, 499.2 KHz	124.8KHz, 499.2KHz	Baud clocks on J3 pins 10 and 12
C0–C15, T1–T3, CLK, DB	C0–C15, T1–T3, CLK, DB	Video timing
HS, VS, CHLD, SVID, COMP SYNC	/HS, /VS, /CHLD, /SVID, COMP_SYNC	Video output
DAC DISABLE, COLOR EN, CD0–CD3	DAC_DISABLE, COLOR_EN, CD0–CD3	Control register, colour
(none)	/RAM_0000 … /RAM_1C00	RAM chip selects by address
(none)	/ROM_A000 … /ROM_B800	BASIC ROM selects by address
(none)	VRAM_A0–A9, VRAM_D0–D7, CG_D0–D7	Video RAM and character generator buses
(none)	KB_ROW0–7, KB_COL0–7, KB_DRV0–7	Keyboard matrix, numbered by data bit
(none)	OSC, OSC_X2, OSC_DIV4, MODE32_GATE	Oscillator and 32/64 column gating
(none)	+5V_IN, POR_CAP, DAC_SUM, /BREAK	Power entry, reset and sound

Design checks

The schematic netlist matches the PCB (269 nets, no differences). DRC reports no violations and no unconnected pads with the default clearance set to 0.15 mm, the original artwork's spacing. Checks that only concern the original artwork, silkscreen and footprint-library differences are set to ignore.